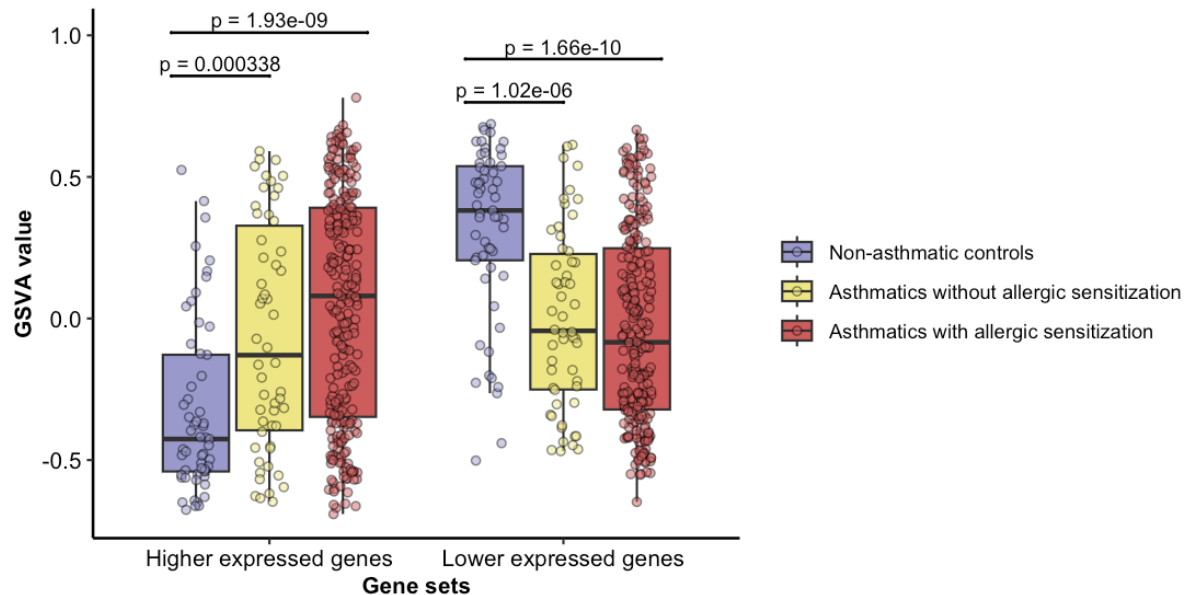
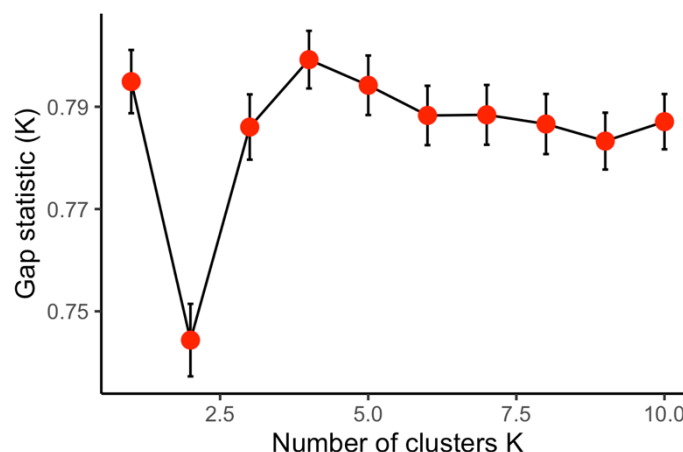


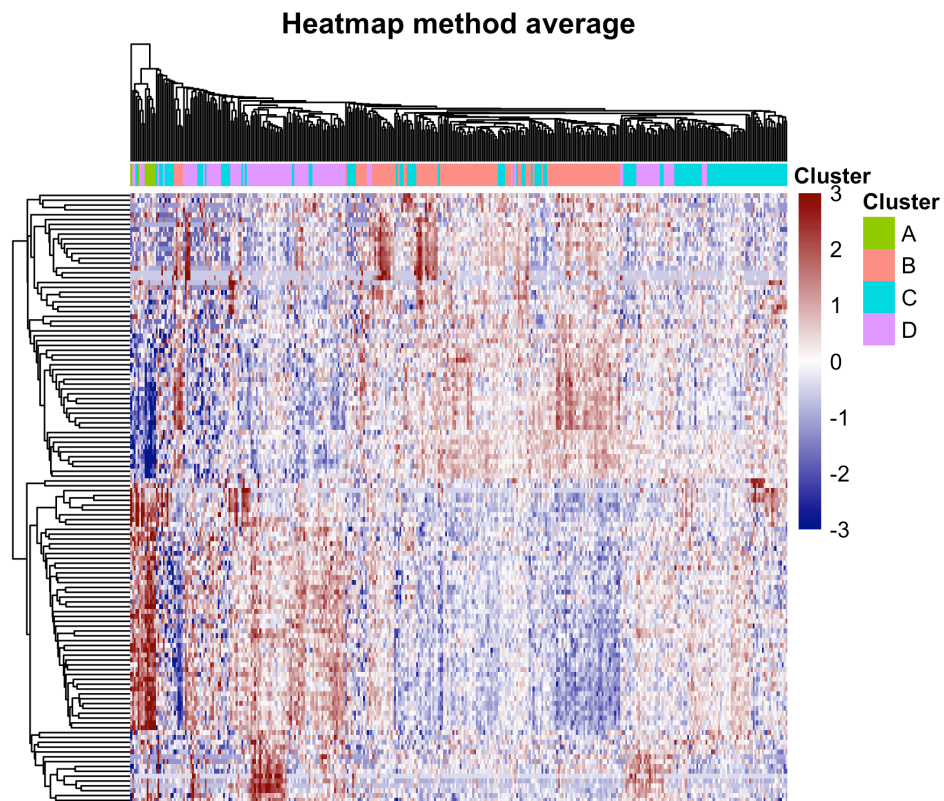
Supplementary material



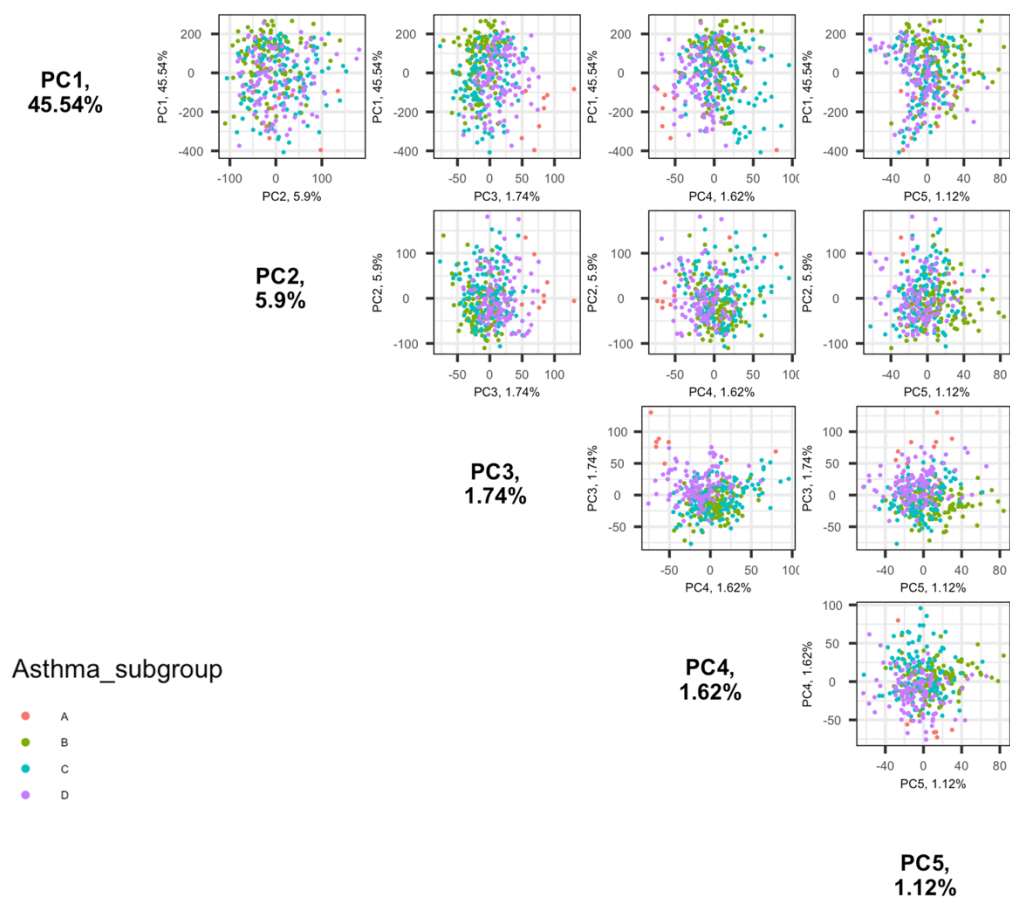
Supplementary Figure 1. Comparison of GSVA scores of higher (67 genes) and lower (59 genes) expressed gene sets identified in ATLANTIS study across participants of the ATLANTIS study. Asthmatic participants are stratified by allergic sensitization status, defined by Phadiatop test. Mann-Whitney U test was used to compare GSVA scores. Phadiatop test information was available for the subset of the ATLANTIS participants: 54 non-allergic asthmatics, 245 asthmatics with allergic sensitization. FDR adjusted p-values are shown at the top.



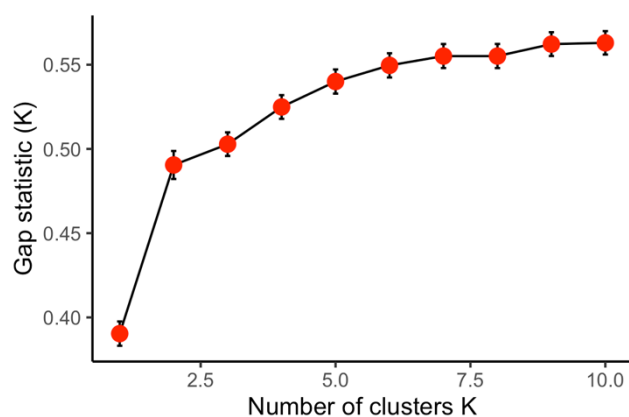
Supplementary Figure 2. Gap statistic to identify optimal number of subgroups for hierarchical clustering of asthma patients. Gap statistic was calculated using clusGap function using 100 bootstrap samples and d.power = 1.



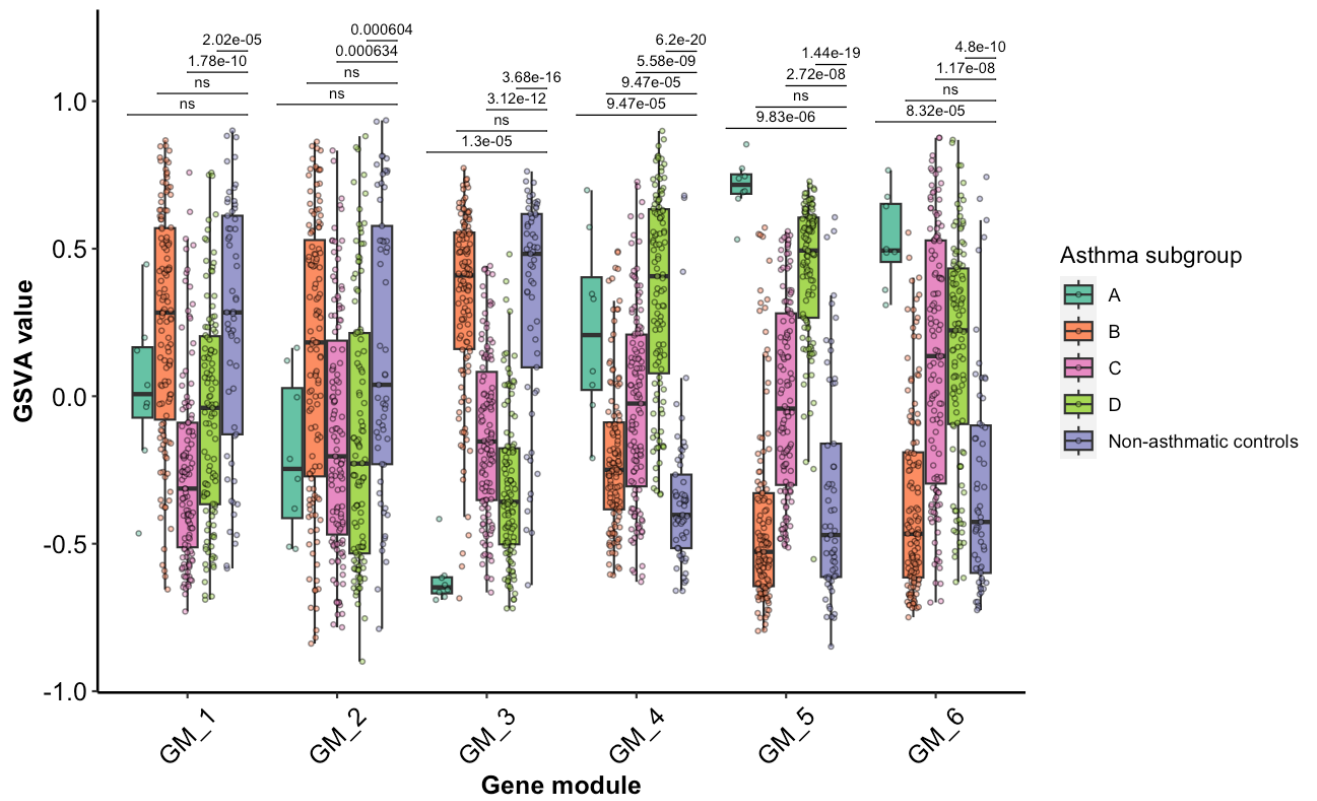
Supplementary Figure 3. Heatmap represents the normalized expression level of 126 differentially expressed genes for asthma patients and non-asthmatic controls. Asthma patients are hierarchically clustered using average linkage method. The color legend represents asthma subgroups identified using complete linkage method.



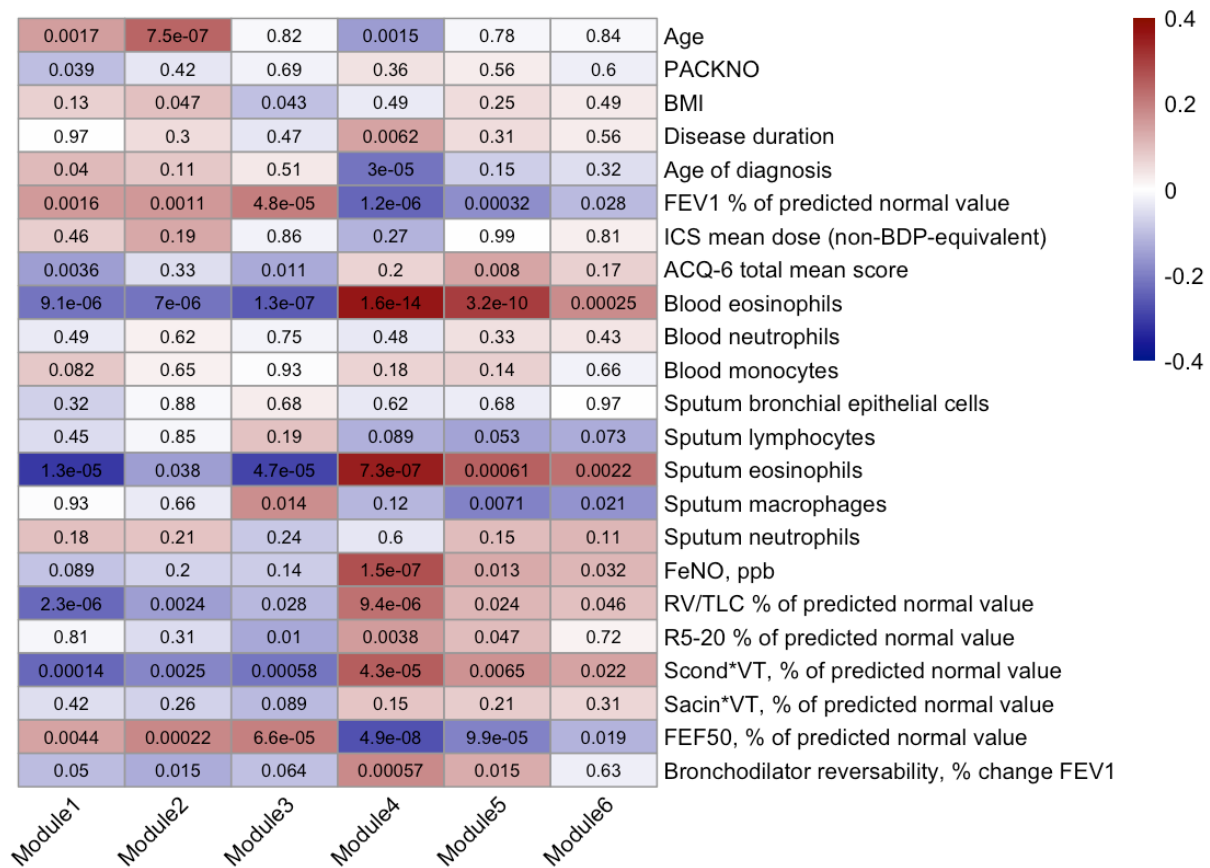
Supplementary Figure 4. PCA analysis on all noticeably expressed genes (genes that have at least 1 CPM in at least 1 sample). Colors represent asthma subgroups. No clear separation of the 4 asthma clusters on the PCA plots.



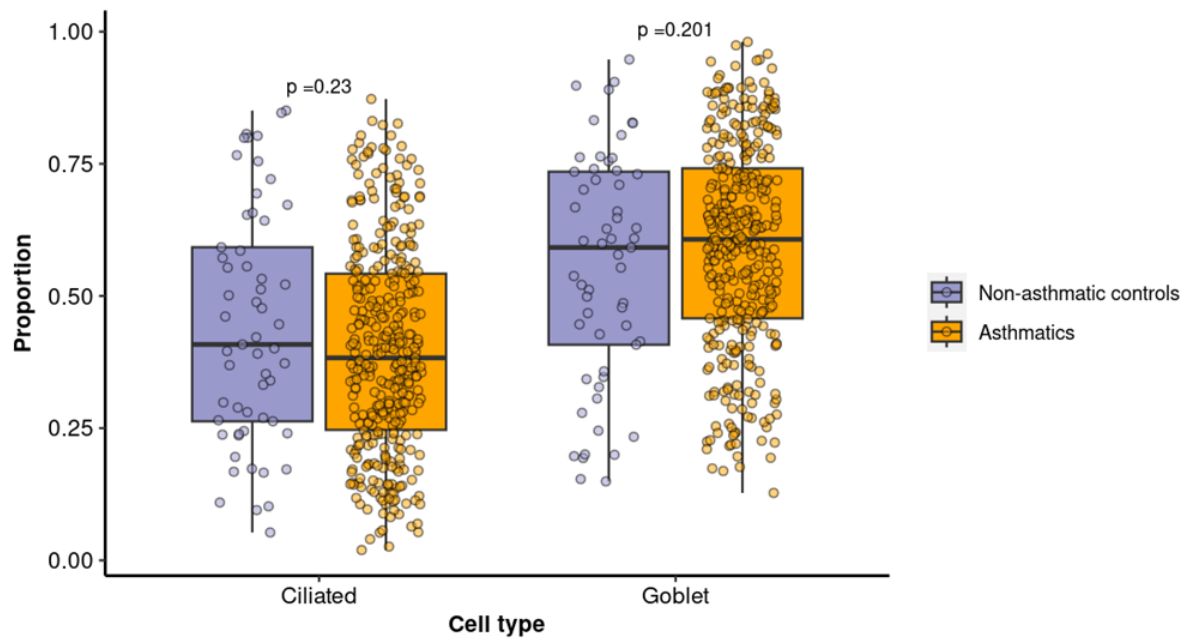
Supplementary Figure 5. Gap statistic to identify optimal number of gene modules for hierarchical clustering of asthma-associated genes. Gap statistic was calculated using `clusGap` function using 100 bootstrap samples and `d.power = 1`. Gap statistic steadily increases with the increase of the number of clusters. Therefore, the number of gene modules was defined by cutting a dendrogram tree and evaluating the resulting clustering visually.



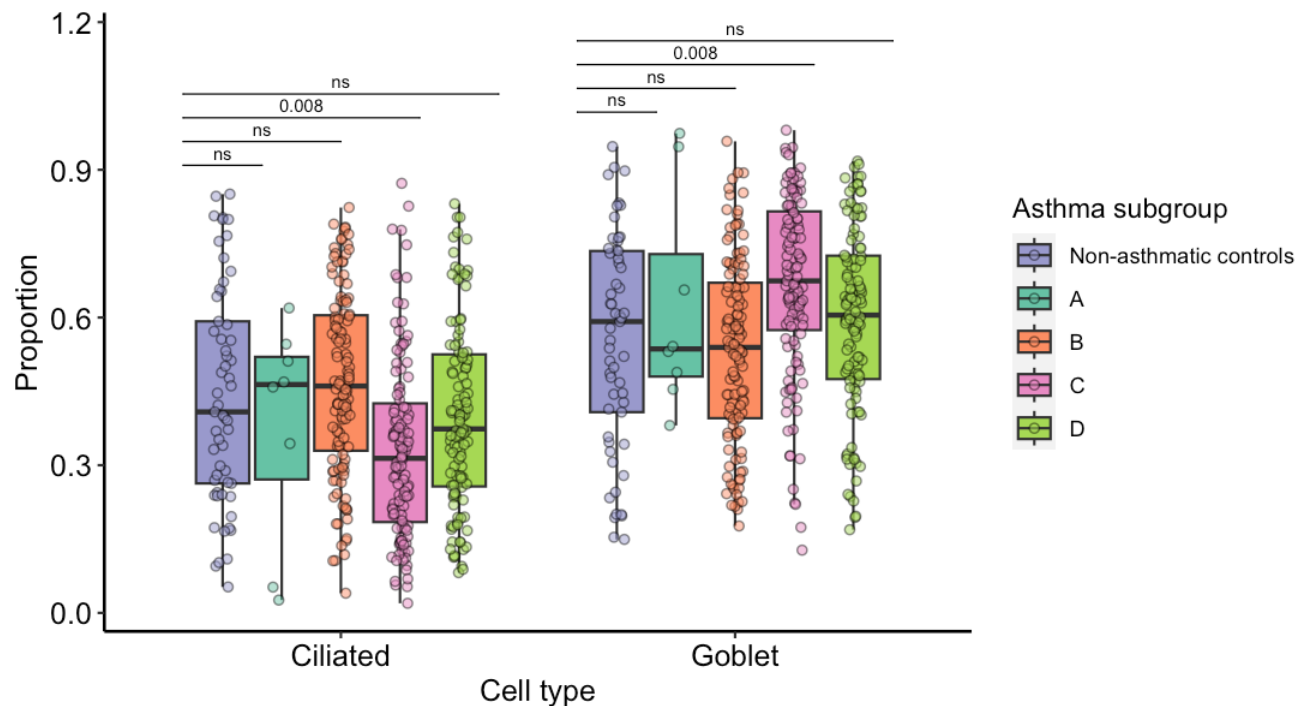
Supplementary Figure 6. Comparison of GSVA scores of 6 gene modules (**Supplementary Table 1**) between 4 defined asthma subgroups and non-asthmatic controls. **GM**: gene module. Comparison was done with Mann-Whitney U test. FDR adjusted p-values are shown at the top, ns: FDR > 0.05.



Supplementary Figure 7. Correlation heatmap between GSVA score of gene modules and clinical characteristics. Color gradient represents the Spearman correlation coefficient, the numbers represent the p-value. **FEV₁**: Forced expiratory volume in 1 second, **ICS**: inhaled corticosteroids, **ACQ-6**: asthma control questionnaire 6, **FeNO**: fractional exhaled nitric oxide, **RV/TLC**: Residual volume/total lung capacity, **R5-20**: resistance at 5 Hz–resistance at 20 Hz, **Scond*VT**: ventilation heterogeneity in the conductive zone of the lungs, **Sacin*VT**: ventilation homogeneity of the acinar zone of the lungs, **FEF50**: Forced expiratory flow at 50% of predicted normal value.



Supplementary Figure 8. Comparison of cell proportions between asthma patients and non-asthmatic controls. Comparison was done with Mann-Whitney U test. P-values are shown at the top of the plot. Analysis was performed with MuSiC cell type deconvolution tool. Normality of data distribution was assessed using histograms.



Supplementary Figure 9. Comparison of cell proportions between asthma subgroups and non-asthmatic controls. Comparison was done with Mann-Whitney U test. FDR adjusted p-values are shown at the top, ns: FDR > 0.05. Analysis was performed with MuSiC cell type deconvolution tool.

	Gene	hgnc_symbol	logFC	PValue	FDR	Gene module	Gene description
1	ENSG00000110436	SLC1A2	-0.935	3.62e-06	0.00555	1	solute carrier family 1 member 2
2	ENSG00000149090	PAMR1	-0.884	0.000304	0.0455	1	peptidase domain containing associated with muscle regeneration 1
3	ENSG00000149021	SCGB1A1	-0.969	3.51e-06	0.00555	1	secretoglobin family 1A member 1
4	ENSG00000150471	ADGRL3	-0.668	0.000333	0.0478	1	adhesion G protein-coupled receptor L3
5	ENSG00000173546	CSPG4	-0.61	0.000173	0.0376	1	chondroitin sulfate proteoglycan 4
6	ENSG00000113721	PDGFRB	-1.02	1.8e-05	0.0133	1	platelet derived growth factor receptor beta
7	ENSG00000139329	LUM	-1.19	2.8e-06	0.00555	1	lumican
8	ENSG00000186340	THBS2	-1.12	4.36e-05	0.0222	1	thrombospondin 2
9	ENSG00000099953	MMP11	-0.687	0.000162	0.0362	1	matrix metalloproteinase 11
10	ENSG00000115414	FN1	-1.03	0.000149	0.0362	1	fibronectin 1
11	ENSG00000011465	DCN	-0.855	0.000246	0.0419	1	decorin
12	ENSG00000087245	MMP2	-0.788	6.11e-05	0.0253	1	matrix metalloproteinase 2
13	ENSG00000113140	SPARC	-0.795	0.00022	0.0412	1	secreted protein acidic and cysteine rich
14	ENSG00000134871	COL4A2	-0.718	1.52e-05	0.0129	1	collagen type IV alpha 2 chain
15	ENSG00000130635	COL5A1	-0.712	5.2e-05	0.0244	1	collagen type V alpha 1 chain
16	ENSG00000163359	COL6A3	-1	2.56e-06	0.00555	1	collagen type VI alpha 3 chain
17	ENSG00000168542	COL3A1	-1.12	6.89e-07	0.00409	1	collagen type III alpha 1 chain
18	ENSG00000164692	COL1A2	-1.33	1.25e-06	0.00447	1	collagen type I alpha 2 chain
19	ENSG00000179477	ALOX12B	-1.04	3.69e-05	0.0212	2	arachidonate 12-lipoxygenase, 12R type
20	ENSG00000138448	ITGAV	-0.19	4.05e-05	0.0219	2	integrin subunit alpha V
21	ENSG00000186395	KRT10	-0.695	5.43e-06	0.00605	2	keratin 10
22	ENSG00000206075	SERPINB5	-0.453	0.00022	0.0412	2	serpin family B member 5
23	ENSG00000167779	IGFBP6	-0.928	2.86e-06	0.00555	2	insulin like growth factor binding protein 6
24	ENSG00000065618	COL17A1	-0.705	3.3e-05	0.0196	2	collagen type XVII alpha 1 chain
25	ENSG00000186847	KRT14	-1.09	0.000136	0.0355	2	keratin 14

26	ENSG00000129116	PALLD	-0.232	5.92e-05	0.0253	3	palladin, cytoskeletal associated protein
27	ENSG00000176731	RBIS	-0.241	0.000269	0.043	3	ribosomal biogenesis factor
28	ENSG00000136824	SMC2	-0.262	7.43e-05	0.0276	3	structural maintenance of chromosomes 2
29	ENSG00000162688	AGL	-0.24	0.000265	0.043	3	amylo-alpha-1, 6-glucosidase, 4-alpha-glucanotransferase
30	ENSG00000158417	EIF5B	-0.184	0.000109	0.0342	3	eukaryotic translation initiation factor 5B
31	ENSG00000135387	CAPRIN1	-0.157	0.000127	0.0352	3	cell cycle associated protein 1
32	ENSG00000165119	HNRNPK	-0.158	0.000286	0.0446	3	heterogeneous nuclear ribonucleoprotein K
33	ENSG00000113013	HSPA9	-0.151	0.000294	0.045	3	heat shock protein family A (Hsp70) member 9
34	ENSG00000233822	H2BC15	-0.373	0.000239	0.0419	3	H2B clustered histone 15
35	ENSG00000147650	LRP12	-0.308	1.24e-05	0.0111	3	LDL receptor related protein 12
36	ENSG00000146842	TMEM209	-0.276	0.00026	0.0426	3	transmembrane protein 209
37	ENSG00000136108	CKAP2	-0.367	1.88e-05	0.0133	3	cytoskeleton associated protein 2
38	ENSG00000138182	KIF20B	-0.329	0.000335	0.0478	3	kinesin family member 20B
39	ENSG00000166428	PLD4	-0.534	0.000129	0.0352	3	phospholipase D family member 4
40	ENSG00000179639	FCER1A	-0.697	3.19e-05	0.0196	3	Fc epsilon receptor 1a
41	ENSG00000116031	CD207	-0.686	9.96e-05	0.0325	3	CD207 molecule
42	ENSG00000143368	SF3B4	-0.204	0.00023	0.0418	3	splicing factor 3b subunit 4
43	ENSG00000117632	STMN1	-0.385	0.000145	0.0362	3	stathmin 1
44	ENSG00000147140	NONO	-0.0929	0.000105	0.0333	3	non-POU domain containing octamer binding
45	ENSG00000196230	TUBB	-0.191	0.000247	0.0419	3	tubulin beta class I
46	ENSG00000070756	PABPC1	-0.151	1.71e-05	0.0132	3	poly(A) binding protein cytoplasmic 1
47	ENSG00000108107	RPL28	-0.191	0.000141	0.0358	3	ribosomal protein L28
48	ENSG00000221983	UBA52	-0.156	0.00021	0.0403	3	ubiquitin A-52 residue ribosomal protein fusion product 1
49	ENSG00000162244	RPL29	-0.29	4.36e-06	0.00597	3	ribosomal protein L29
50	ENSG00000198918	RPL39	-0.258	0.000126	0.0352	3	ribosomal protein L39
51	ENSG00000188846	RPL14	-0.186	6.93e-05	0.027	3	ribosomal protein L14

52	ENSG00000144713	RPL32	-0.261	7.95e-06	0.00833	3	ribosomal protein L32
53	ENSG00000174444	RPL4	-0.197	0.000131	0.0354	3	ribosomal protein L4
54	ENSG00000065978	YBX1	-0.203	7.13e-05	0.027	3	Y-box binding protein 1
55	ENSG00000114942	EEF1B2	-0.214	0.000261	0.0426	3	eukaryotic translation elongation factor 1 beta 2
56	ENSG00000181163	NPM1	-0.237	3.74e-06	0.00555	3	nucleophosmin 1
57	ENSG00000145741	BTF3	-0.184	0.000125	0.0352	3	basic transcription factor 3
58	ENSG00000171490	RSL1D1	-0.201	9.38e-05	0.0315	3	ribosomal L1 domain containing 1
59	ENSG00000163468	CCT3	-0.162	0.000189	0.0379	3	chaperonin containing TCP1 subunit 3
60	ENSG00000188559	RALGAPA2	0.143	0.00016	0.0362	4	Ral GTPase activating protein catalytic subunit alpha 2
61	ENSG00000130821	SLC6A8	0.421	0.000133	0.0355	4	solute carrier family 6 member 8
62	ENSG00000117266	CDK18	0.27	0.000124	0.0352	4	cyclin dependent kinase 18
63	ENSG00000144791	LIMD1	0.149	0.000159	0.0362	4	LIM domain containing 1
64	ENSG00000175538	KCNE3	0.276	7.71e-05	0.028	4	potassium voltage-gated channel subfamily E regulatory subunit 3
65	ENSG00000256612	CYP2B7P	0.346	0.000122	0.0352	4	cytochrome P450 family 2 subfamily B member 7, pseudogene
66	ENSG00000185274	GALNT17	0.761	0.000296	0.045	4	polypeptide N-acetylgalactosaminyltransferase 17
67	ENSG00000016490	CLCA1	3.81	5.03e-10	8.95e-06	4	chloride channel accessory 1
68	ENSG00000090512	FETUB	2.26	3.22e-07	0.00287	4	fetuin B
69	ENSG00000124215	CDH26	0.723	1.08e-06	0.00447	4	cadherin 26
70	ENSG00000170373	CST1	2.31	1.01e-05	0.01	4	cystatin SN
71	ENSG00000274114	ALOX15P1	0.824	5.07e-06	0.00605	4	arachidonate 15-lipoxygenase pseudogene 1
72	ENSG00000183196	CHST6	0.267	0.000165	0.0362	4	carbohydrate sulfotransferase 6
73	ENSG00000286811	NA	0.82	0.00019	0.0379	4	NA
74	ENSG00000168621	GDNF	1.11	3.96e-05	0.0219	4	glial cell derived neurotrophic factor
75	ENSG00000148053	NTRK2	0.961	1.19e-05	0.0111	4	neurotrophic receptor tyrosine kinase 2

76	ENSG00000133110	POSTN	1.15	0.000252	0.0422	4	periostin
77	ENSG00000132793	LPIN3	0.273	1.00e-04	0.0325	5	lipin 3
78	ENSG00000139631	CSAD	0.308	2.09e-05	0.0138	5	cysteine sulfinic acid decarboxylase
79	ENSG00000221978	CCNL2	0.194	4.83e-05	0.0232	5	cyclin L2
80	ENSG00000196123	KIAA0895L	0.271	0.000245	0.0419	5	microtubule associated tyrosine carboxypeptidase 1
81	ENSG00000102878	HSF4	0.407	0.000273	0.043	5	heat shock transcription factor 4
82	ENSG00000169246	NPIPB3	0.346	0.000198	0.0392	5	nuclear pore complex interacting protein family member B3
83	ENSG00000258839	MC1R	0.504	5.89e-05	0.0253	5	melanocortin 1 receptor
84	ENSG00000130787	HIP1R	0.239	0.000117	0.0352	5	huntingtin interacting protein 1 related
85	ENSG00000205356	TECPR1	0.24	0.000229	0.0418	5	tectonin beta-propeller repeat containing 1
86	ENSG00000130158	DOCK6	0.24	0.000205	0.04	5	dedicator of cytokinesis 6
87	ENSG00000197070	ARRDC1	0.255	0.000288	0.0446	5	arrestin domain containing 1
88	ENSG00000163945	UVSSA	0.319	5.12e-06	0.00605	5	UV stimulated scaffold protein A
89	ENSG00000167615	LENG8	0.267	0.000257	0.0426	5	leukocyte receptor cluster member 8
90	ENSG00000164877	MICALL2	0.318	0.000235	0.0419	5	MICAL like 2
91	ENSG00000180104	EXOC3	0.19	0.000298	0.045	5	exocyst complex component 3
92	ENSG00000243679	NA	0.522	0.000165	0.0362	5	NA
93	ENSG00000129566	TEP1	0.233	4.6e-05	0.0227	5	telomerase associated protein 1
94	ENSG00000110719	TCIRG1	0.358	0.00019	0.0379	5	T cell immune regulator 1, ATPase H ⁺ transporting V0 subunit a3
95	ENSG00000130827	PLXNA3	0.287	5.67e-05	0.0253	5	plexin A3
96	ENSG00000104365	IKBKB	0.168	0.000164	0.0362	5	inhibitor of nuclear factor kappa B kinase subunit beta
97	ENSG00000104957	CCDC130	0.209	0.000237	0.0419	5	YJU2 splicing factor homolog B
98	ENSG00000105321	CCDC9	0.422	1.68e-05	0.0132	5	coiled-coil domain containing 9
99	ENSG00000166987	MBD6	0.274	0.000327	0.0477	5	methyl-CpG binding domain protein 6
100	ENSG00000129925	PGAP6	0.219	0.000185	0.0379	5	post-GPI attachment to proteins 6

101	ENSG00000064687	ABCA7	0.36	0.000187	0.0379	5	ATP binding cassette subfamily A member 7
102	ENSG00000257594	GALNT4	0.551	8.85e-05	0.0303	5	polypeptide N-acetylgalactosaminyltransferase 4
103	ENSG00000166126	AMN	0.473	0.000188	0.0379	5	amnion associated transmembrane protein
104	ENSG00000279476	NA	0.424	0.000227	0.0418	5	NA
105	ENSG00000225138	SLC9A3-AS1	0.48	0.000137	0.0355	5	SLC9A3 antisense RNA 1
106	ENSG00000112877	CEP72	0.462	0.000307	0.0455	5	centrosomal protein 72
107	ENSG00000117616	RSRP1	0.209	4.27e-05	0.0222	5	arginine and serine rich protein 1
108	ENSG00000189339	SLC35E2B	0.21	0.000336	0.0478	5	solute carrier family 35 member E2B
109	ENSG00000237441	RGL2	0.204	6.94e-05	0.027	5	ral guanine nucleotide dissociation stimulator like 2
110	ENSG00000112511	PHF1	0.201	0.000157	0.0362	5	PHD finger protein 1
111	ENSG00000179698	WDR97	0.57	0.000156	0.0362	5	WD repeat domain 97
112	ENSG00000198089	SFI1	0.259	1.94e-05	0.0133	5	SFI1 centrin binding protein
113	ENSG00000196187	TMEM63A	0.194	0.000189	0.0379	5	transmembrane protein 63A
114	ENSG00000168395	ING5	0.219	0.000341	0.0482	5	inhibitor of growth family member 5
115	ENSG00000126217	MCF2L	0.29	8.13e-05	0.0284	5	MCF.2 cell line derived transforming sequence like
116	ENSG00000187609	EXD3	0.358	8.14e-05	0.0284	5	exonuclease 3'-5' domain containing 3
117	ENSG00000279602	NA	1.08	2.66e-05	0.0169	6	NA
118	ENSG00000177606	JUN	0.446	0.000116	0.0352	6	Jun proto-oncogene, AP-1 transcription factor subunit
119	ENSG0000022567	SLC45A4	0.379	2.09e-06	0.00555	6	solute carrier family 45 member 4
120	ENSG00000129657	SEC14L1	0.303	7.00e-05	0.027	6	SEC14 like lipid binding 1
121	ENSG00000167703	SLC43A2	0.35	6.09e-05	0.0253	6	solute carrier family 43 member 2
122	ENSG00000142102	PGGHG	0.623	0.000207	0.04	6	protein-glucosylgalactosylhydroxylysine glucosidase
123	ENSG00000267519	NA	0.737	0.00027	0.043	6	miR-23a/27a/24-2 cluster host gene
124	ENSG00000287937	NA	1.56	0.00015	0.0362	6	uncharacterized LOC124907871

125	ENSG00000170345	FOS	1.11	0.000242	0.0419	6	Fos proto-oncogene, AP-1 transcription factor subunit
126	ENSG00000125740	FOSB	1.13	0.000309	0.0455	6	FosB proto-oncogene, AP-1 transcription factor subunit

Supplementary Table 1. List of the differentially expressed genes between asthma patients and non-asthmatic controls identified in the ATLANTIS study. Multiple testing correction was performed using Benjamini-Hochberg procedure. log₂FC: log₂ fold change, FDR: false discovery rate. Genes are ordered by the module membership.

Ensembl gene ID	hgnc_symbol	log ₂ FC	p-value	FDR
ENSG00000016490	CLCA1	3.26	1.91e-07	4.02e-06
ENSG00000170373	CST1	2.15	2.25e-05	0.000102
ENSG00000090512	FETUB	1.82	9.8e-06	7.08e-05
ENSG00000287937	-	1.12	0.0096	0.0096
ENSG00000170345	FOS	1.1	5.42e-05	0.00016
ENSG00000279602	-	0.991	6.34e-06	5.33e-05
ENSG00000125740	FOSB	0.88	0.00115	0.00137
ENSG00000133110	POSTN	0.878	0.00342	0.00359
ENSG00000148053	NTRK2	0.802	0.000191	0.000404
ENSG00000274114	ALOX15P1	0.728	1.78e-05	9.98e-05
ENSG00000168621	GDNF	0.689	0.00239	0.00269
ENSG00000267519	MIR23AHG	0.673	0.000387	0.000676
ENSG00000185274	GALNT17	0.668	0.000516	0.000793

ENSG00000124215	CDH26	0.604	3.94e-05	0.000138
ENSG00000286811	-	0.597	0.00281	0.003
ENSG00000257594	GALNT4	0.554	4.46e-05	0.000144
ENSG00000142102	PGGHG	0.55	0.000482	0.000769
ENSG00000179698	WDR97	0.513	0.000304	0.000564
ENSG00000177606	JUN	0.459	2.33e-05	0.000102
ENSG00000243679	-	0.456	0.000743	0.00101
ENSG00000225138	SLC9A3- AS1	0.426	0.000472	0.000762
ENSG00000130821	SLC6A8	0.41	0.000149	0.000342
ENSG00000112877	CEP72	0.406	0.00117	0.00137
ENSG00000166126	AMN	0.406	0.000783	0.00103
ENSG00000258839	MC1R	0.394	0.000762	0.00101
ENSG00000167703	SLC43A2	0.374	1.98e-05	9.98e-05
ENSG00000022567	SLC45A4	0.344	1.51e-05	9.41e-05
ENSG00000110719	TCIRG1	0.331	0.000355	0.000639
ENSG00000105321	CCDC9	0.33	0.000442	0.000742
ENSG00000064687	ABCA7	0.321	0.000847	0.00109
ENSG00000102878	HSF4	0.316	0.00275	0.00297
ENSG00000256612	CYP2B7P	0.315	0.000373	0.000662
ENSG00000187609	EXD3	0.308	0.000439	0.000742

ENSG00000129657	SEC14L1	0.295	0.000104	0.000263
ENSG00000279476	-	0.288	0.00725	0.00731
ENSG00000163945	UVSSA	0.286	3.04e-05	0.00012
ENSG00000175538	KCNE3	0.276	5.37e-05	0.00016
ENSG00000164877	MICALL2	0.264	0.00206	0.00234
ENSG00000117266	CDK18	0.255	0.000184	0.000399
ENSG00000139631	CSAD	0.253	0.000287	0.000546
ENSG00000126217	MCF2L	0.244	0.000853	0.00109
ENSG00000169246	NPIP3	0.242	0.00682	0.00693
ENSG00000198089	SFI1	0.238	5.7e-05	0.000163
ENSG00000130827	PLXNA3	0.238	0.000756	0.00101
ENSG00000183196	CHST6	0.238	0.000551	0.000826
ENSG00000129925	PGAP6	0.234	6.53e-05	0.000183
ENSG00000132793	LPIN3	0.228	0.000655	0.000916
ENSG00000205356	TECPR1	0.224	0.000538	0.000817
ENSG00000129566	TEP1	0.223	9.45e-05	0.000248
ENSG00000166987	MBD6	0.222	0.00261	0.00285
ENSG00000197070	ARRDC1	0.222	0.00116	0.00137
ENSG00000167615	LENG8	0.213	0.00294	0.00311
ENSG00000237441	RGL2	0.205	4.44e-05	0.000144
ENSG00000130787	HIP1R	0.205	0.00084	0.00109

ENSG00000196123	MATCAP1	0.204	0.0035	0.00365
ENSG00000130158	DOCK6	0.201	0.00167	0.00192
ENSG00000189339	SLC35E2B	0.2	0.000513	0.000793
ENSG00000104957	YJU2B	0.194	0.000463	0.000757
ENSG00000112511	PHF1	0.192	0.000247	0.000479
ENSG00000168395	ING5	0.191	0.000941	0.00117
ENSG00000117616	RSRP1	0.176	0.000567	0.000841
ENSG00000221978	CCNL2	0.176	0.000204	0.000415
ENSG00000180104	EXOC3	0.172	0.000889	0.00112
ENSG00000188559	RALGAPA2	0.166	1.95e-05	9.98e-05
ENSG00000144791	LIMD1	0.155	0.000127	0.000303
ENSG00000196187	TMEM63A	0.153	0.00251	0.0028
ENSG00000104365	IKBKB	0.123	0.00411	0.00425
ENSG00000147140	NONO	-0.0838	0.000291	0.000546
ENSG00000070756	PABPC1	-0.128	0.000193	0.000404
ENSG00000165119	HNRNPK	-0.13	0.00259	0.00285
ENSG00000113013	HSPA9	-0.136	0.000971	0.0012
ENSG00000135387	CAPRIN1	-0.136	0.000753	0.00101
ENSG00000221983	UBA52	-0.14	0.000705	0.000976
ENSG00000163468	CCT3	-0.163	0.000124	0.000301
ENSG00000108107	RPL28	-0.172	0.000338	0.000618

ENSG00000174444	RPL4	-0.175	0.000494	0.000778
ENSG00000143368	SF3B4	-0.175	0.000634	0.000898
ENSG00000145741	BTF3	-0.176	0.000184	0.000399
ENSG00000158417	EIF5B	-0.183	5.46e-05	0.00016
ENSG00000188846	RPL14	-0.184	8.07e-05	0.000216
ENSG00000196230	TUBB	-0.186	0.000164	0.000369
ENSG00000171490	RSL1D1	-0.186	0.000124	0.000301
ENSG00000065978	YBX1	-0.19	0.000134	0.000313
ENSG00000138448	ITGAV	-0.191	2.44e-05	0.000102
ENSG00000114942	EEF1B2	-0.199	0.000406	0.000701
ENSG00000146842	TMEM209	-0.202	0.00276	0.00297
ENSG00000198918	RPL39	-0.212	0.00123	0.00142
ENSG00000129116	PALLD	-0.213	0.000228	0.000449
ENSG00000176731	RBIS	-0.22	0.000228	0.000449
ENSG00000162688	AGL	-0.221	0.000627	0.000898
ENSG00000144713	RPL32	-0.233	5.36e-05	0.00016
ENSG00000181163	NPM1	-0.237	2.58e-06	2.7e-05
ENSG00000136824	SMC2	-0.239	0.000104	0.000263
ENSG00000162244	RPL29	-0.254	3.55e-05	0.000132
ENSG00000138182	KIF20B	-0.294	2,00E-04	0.000413
ENSG00000147650	LRP12	-0.301	1.01e-05	7.08e-05

ENSG00000233822	H2BC15	-0.304	0.000594	0.00087
ENSG00000136108	CKAP2	-0.328	3.36e-05	0.000128
ENSG00000117632	STMN1	-0.379	1.57e-05	9.41e-05
ENSG00000166428	PLD4	-0.469	7.95e-05	0.000216
ENSG00000206075	SERPINB5	-0.497	1.9e-05	9.98e-05
ENSG00000116031	CD207	-0.503	0.000628	0.000898
ENSG00000110436	SLC1A2	-0.525	0.00104	0.00127
ENSG00000150471	ADGRL3	-0.526	0.000454	0.000752
ENSG00000149090	PAMR1	-0.533	0.00492	0.00504
ENSG00000179639	FCER1A	-0.585	2.5e-05	0.000102
ENSG00000099953	MMP11	-0.636	3.94e-05	0.000138
ENSG00000065618	COL17A1	-0.656	2.36e-05	0.000102
ENSG00000186395	KRT10	-0.693	5.64e-07	7.89e-06
ENSG00000130635	COL5A1	-0.733	3,00E-06	2.91e-05
ENSG00000113140	SPARC	-0.763	4.19e-05	0.000143
ENSG00000149021	SCGB1A1	-0.766	2.47e-05	0.000102
ENSG00000134871	COL4A2	-0.79	2.2e-08	6.92e-07
ENSG00000173546	CSPG4	-0.793	1.76e-08	6.92e-07
ENSG00000087245	MMP2	-0.796	3.57e-06	3.22e-05
ENSG00000186847	KRT14	-0.848	0.00122	0.00142
ENSG00000011465	DCN	-0.856	1.46e-05	9.41e-05

ENSG00000115414	FN1	-0.866	0.00104	0.00127
ENSG00000163359	COL6A3	-0.947	1.92e-06	2.2e-05
ENSG00000168542	COL3A1	-1.1	2.68e-07	4.82e-06
ENSG00000113721	PDGFRB	-1.11	9.04e-08	2.28e-06
ENSG00000186340	THBS2	-1.15	7.84e-06	6.17e-05
ENSG00000139329	LUM	-1.23	5.14e-07	7.89e-06
ENSG00000167779	IGFBP6	-1.25	3.76e-14	4.74e-12
ENSG00000164692	COL1A2	-1.26	1.08e-06	1.36e-05
ENSG00000179477	ALOX12B	-1.51	3.35e-11	2.11e-09

Supplementary Table 2. List of 126 asthma-associated genes after sensitivity analysis adjusting for study site (using model with extra 16 dummy variables to account for 17 different study sites). Multiple testing correction was performed for 126 genes using Benjamini-Hochberg procedure. log₂FC: log₂ fold change, FDR: false discovery rate.

	A	rest (B,C,D)	p-val	B	rest (A,C,D)	p-val	C	rest (A,B,D)	p-val	D	rest (A,B,C)	p-val
n	8	353		123	238		119	242		111	250	
Sex , male, n(%)	1 (12.5)	148 (41.9)	0.19	51 (41.5)	98 (41.2)	1	44 (37.0)	105 (43.4)	0.29	53 (47.7)	96 (38.4)	0.12
Age , years (mean (SD))	40.5 (17.6)	43.9 (13.1)	0.47	44.6 (12.8)	43.5 (13.4)	0.46	43.9 (13.1)	43.8 (13.3)	0.95	43.2 (13.6)	44.1 (13.1)	0.56
Smoking status , n(%)			0.83			0.91			0.06			0.19
Current-smoker	0 (0.0)	16 (4.5)		5 (4.1)	11 (4.6)		9 (7.6)	7 (2.9)		2 (1.8)	14 (5.6)	
Ex-smoker	2 (25.0)	84 (23.8)		28 (22.8)	58 (24.4)		32 (26.9)	54 (22.3)		24 (21.6)	62 (24.8)	
Non-smoker	6 (75.0)	253 (71.7)		90 (73.2)	169 (71.0)		78 (65.5)	181 (74.8)		85 (76.6)	174 (69.6)	
BMI , kg/m ² (mean (SD))	27.5 (6.1)	27.0 (5.4)	0.81	26.5 (4.9)	27.3 (5.7)	0.22	27.0 (5.4)	27.1 (5.5)	0.89	27.6 (6.0)	26.8 (5.2)	0.2
Phadiatop test , Positive, n(%)	6 (85.7)	239 (81.8)	1	83 (76.1)	162 (85.3)	0.07	81 (87.1)	164 (79.6)	0.16	75 (83.3)	170 (81.3)	0.81
Duration of disease , years (median (IQR))	17.8 [12.3, 21.2]	20.6 [9.0, 31.1]	0.31	19.1 [7.6, 29.8]	20.7 [10.9, 31.0]	0.36	20.1 [10.7, 30.4]	20.7 [8.1, 31.0]	0.91	21.7 [10.9, 32.5]	19.9 [8.6, 29.6]	0.25

Age at first diagnosis, years (median (IQR))	22.0 [5.5, 42.5]	20.4 [6.7, 36.0]	0.82	23.0 [10.0, 38.5]	18.7 [6.0, 35.0]	0.07	20.7 [6.8, 32.8]	20.2 [6.3, 36.8]	0.84	15.8 [5.6, 35.5]	21.1 [8.0, 36.2]	0.08
FEV₁, % predicted (mean (SD))	93.8 (15.4)	82.8 (17.3)	0.08	85.9 (16.8)	81.6 (17.4)	0.03	81.3 (17.3)	83.9 (17.3)	0.19	81.0 (17.5)	84.0 (17.2)	0.14
ACQ-6 total mean score (median (IQR))	1.3 [0.5, 2.0]	0.8 [0.3, 1.5]	0.44	0.7 [0.3, 1.2]	0.8 [0.3, 1.7]	0.04	1.0 [0.3, 1.6]	0.8 [0.3, 1.3]	0.34	0.8 [0.3, 1.7]	0.8 [0.3, 1.5]	0.35
Inhaled corticosteroid usage, n (%)	4 (50.0)	91 (25.8)	0.26	33 (26.8)	62 (26.1)	0.97	29 (24.4)	66 (27.3)	0.64	29 (26.1)	66 (26.4)	1
Blood eosinophils, 10⁹/L (median (IQR))	0.3 [0.1, 0.5]	0.2 [0.1, 0.3]	0.45	0.2 [0.1, 0.3]	0.2 [0.2, 0.4]	<0.001	0.2 [0.1, 0.3]	0.2 [0.1, 0.4]	0.87	0.3 [0.2, 0.4]	0.2 [0.1, 0.3]	<0.001
Sputum lymphocytes, % of non-squamous cells (median (IQR))	0.8 [0.0, 2.0]	0.7 [0.3, 1.7]	0.73	0.8 [0.3, 1.7]	0.7 [0.3, 1.7]	0.81	0.9 [0.5, 2.1]	0.6 [0.3, 1.6]	0.02	0.4 [0.3, 1.3]	0.8 [0.4, 1.9]	0.06
Sputum eosinophils, % of non-squamous cells (median (IQR))	0.4 [0.0, 0.4]	0.5 [0.0, 3.1]	0.70	0.2 [0.0, 0.9]	0.6 [0.1, 4.4]	0.01	0.5 [0.1, 3.0]	0.4 [0.0, 2.9]	0.53	1.2 [0.0, 6.5]	0.3 [0.0, 1.9]	0.03
FeNO, ppb (median (IQR))	18.0 [7.0, 23.0]	24.0 [16.0, 38.3]	0.10	24.0 [15.0, 36.0]	23.0 [16.0, 39.5]	0.74	21.0 [15.8, 30.8]	25.0 [16.0, 41.0]	0.04	29.0 [19.0, 51.0]	22.0 [15.0, 33.8]	0.004
RV/TLC % predicted (mean (SD))	91.7 (16.0)	105.9 (26.9)	0.14	100.4 (24.1)	108.2 (27.7)	0.01	109.7 (28.3)	103.5 (25.8)	0.04	107.9 (27.5)	104.6 (26.4)	0.30
R5-20 % predicted (median (IQR))	455.2 [307.3, 484.2]	264.0 [90.2, 628.4]	0.46	200.2 [69.1, 669.4]	294.2 [95.0, 617.0]	0.43	363.6 [136.9, 645.8]	209.9 [72.7, 605.8]	0.05	194.8 [73.2, 559.7]	294.2 [95.0, 651.6]	0.17
Scond*VT, % predicted (median (IQR))	109.6 [76.0, 132.8]	96.9 [52.2, 160.7]	0.68	77.0 [39.5, 160.4]	103.4 [60.0, 160.0]	0.08	105.9 [58.8, 160.8]	91.7 [46.8, 158.8]	0.27	101.6 [61.9, 160.3]	96.4 [52.2, 160.6]	0.53
Sacin*VT, % predicted (median (IQR))	145.3 [139.0, 186.6]	119.5 [85.9, 171.4]	0.43	125.8 [86.9, 171.4]	118.7 [85.9, 172.3]	0.59	116.3 [92.2, 169.8]	125.8 [83.0, 172.3]	0.91	113.9 [79.7, 172.6]	120.0 [90.8, 171.2]	0.34
FEF50, % predicted (mean (SD))	79.9 (31.8)	67.3 (29.3)	0.23	72.3 (31.8)	65.2 (27.8)	0.04	66.3 (27.8)	68.2 (30.1)	0.59	62.9 (27.4)	69.7 (30.0)	0.05
Bronchodilator reversibility, % change FEV₁	9.0 (3.8)	13.19 (12.5)	0.36	11.5 (11.6)	13.8 (12.7)	0.11	13.4 (13.6)	12.8 (11.8)	0.66	14.5 (12.12)	12.33 (12.47)	0.14
Airway hyper-responsiveness, n (%)			0.816			0.18			0.14			0.01
Very mild	1 (14.3)	70 (27.5)		28 (29.5)	43 (25.7)		30 (36.1)	41 (22.9)		12 (15.6)	59 (31.9)	

Mild	2 (28.6)	72 (28.2)		30 (31.6)	44 (26.3)		21 (25.3)	53 (29.6)		21 (27.3)	53 (28.6)	
Moderate	3 (42.9)	72 (28.2)		28 (29.5)	47 (28.1)		19 (22.9)	56 (31.3)		25 (32.5)	50 (27.0)	
Severe	1 (14.3)	41 (16.1)		9 (9.5)	33 (19.8)		13 (15.7)	29 (16.2)		19 (24.7)	23 (12.4)	

Supplementary Table 3. Comparison of each asthma subgroup with the rest of the asthma patients. **FEV₁**: Forced expiratory volume in 1 second, **FeNO**: fractional exhaled nitric oxide, **RV/TLC**: Residual volume/total lung capacity, **R5-20**: resistance at 5 Hz–resistance at 20 Hz, **Scond*VT**: ventilation heterogeneity in the conductive zone of the lungs, **Sacin*VT**: ventilation homogeneity of the acinar zone of the lungs, **FEF50**: Forced expiratory flow at 50% of predicted normal value. For normal data t-test was used, for non-normal data Mann-Whitney U test was performed. Significant differences (p-value < 0.05) are colored in blue.

	A	B	p	A	C	p	A	D	p
n	8	123		8	119		8	111	
Sex, male, n(%)	1 (12.5)	51 (41.5)	0.211	1 (12.5)	44 (37.0)	0.308	1 (12.5)	53 (47.7)	0.117
Age, years (mean (SD))	40.50 (17.61)	44.56 (12.83)	0.398	40.50 (17.61)	43.91 (13.06)	0.486	40.50 (17.61)	43.24 (13.55)	0.589
Smoking status, n(%)			0.842			0.699			0.911
Current smoker	0 (0.0)	5 (4.1)		0 (0.0)	9 (7.6)		0 (0.0)	2 (1.8)	
Ex smoker	2 (25.0)	28 (22.8)		2 (25.0)	32 (26.9)		2 (25.0)	24 (21.6)	
Non smoker	6 (75.0)	90 (73.2)		6 (75.0)	78 (65.5)		6 (75.0)	85 (76.6)	
BMI, kg/m ² (mean (SD))	27.48 (6.11)	26.54 (4.93)	0.609	27.48 (6.11)	26.97 (5.40)	0.799	27.48 (6.11)	27.60 (5.97)	0.956
Phadiatop test, Positive, n(%)	6 (85.7)	83 (76.1)	0.905	6 (85.7)	81 (87.1)	1.000	6 (85.7)	75 (83.3)	1.000
Duration of disease, years (median (IQR))	17.81 [12.33, 21.24]	19.06 [7.59, 29.79]	0.436	17.81 [12.33, 21.24]	20.11 [10.69, 30.38]	0.336	17.81 [12.33, 21.24]	21.67 [10.88, 32.48]	0.207
Age at first diagnosis, years (median (IQR))	22.02 [5.47, 42.45]	23.00 [10.00, 38.47]	0.916	22.02 [5.47, 42.45]	20.68 [6.79, 32.80]	0.787	22.02 [5.47, 42.45]	15.84 [5.59, 35.50]	0.603
FEV ₁ , % of predicted normal value (mean (SD))	93.80 (15.43)	85.89 (16.84)	0.199	93.80 (15.43)	81.28 (17.27)	0.049	93.80 (15.43)	80.98 (17.53)	0.047
ACQ-6 total mean score (median (IQR))	1.33 [0.46, 2.04]	0.67 [0.33, 1.17]	0.245	1.33 [0.46, 2.04]	1.00 [0.33, 1.58]	0.588	1.33 [0.46, 2.04]	0.83 [0.33, 1.67]	0.602
Blood eosinophils, 10 ⁹ /L (median (IQR))	0.32 [0.10, 0.54]	0.15 [0.10, 0.26]	0.181	0.32 [0.10, 0.54]	0.20 [0.13, 0.30]	0.388	0.32 [0.10, 0.54]	0.29 [0.16, 0.40]	0.944

Sputum lymphocytes, % of non-squamous cells (median (IQR))	0.80 [0.00, 2.00]	0.80 [0.30, 1.65]	0.773	0.80 [0.00, 2.00]	0.90 [0.45, 2.05]	0.446	0.80 [0.00, 2.00]	0.40 [0.30, 1.30]	0.976
Sputum eosinophils, % of non-squamous cells (median (IQR))	0.40 [0.00, 0.40]	0.20 [0.00, 0.90]	0.949	0.40 [0.00, 0.40]	0.50 [0.10, 2.95]	0.563	0.40 [0.00, 0.40]	1.20 [0.00, 6.50]	0.497
FeNO, ppb (median (IQR))	18.00 [7.00, 23.00]	24.00 [15.00, 36.00]	0.115	18.00 [7.00, 23.00]	21.00 [15.75, 30.75]	0.240	18.00 [7.00, 23.00]	29.00 [19.00, 51.00]	0.037
RV/TLC % of predicted normal value (mean (SD))	91.71 (15.97)	100.37 (24.12)	0.320	91.71 (15.97)	109.73 (28.25)	0.077	91.71 (15.97)	107.85 (27.46)	0.104
R5-20 % of predicted normal value (median (IQR))	455.23 [307.33, 484.15]	200.21 [69.14, 669.36]	0.428	455.23 [307.33, 484.15]	363.57 [135.94, 645.84]	0.797	455.23 [307.33, 484.15]	194.84 [73.21, 559.74]	0.285
Scond*VT, % of predicted normal value (median (IQR))	109.61 [76.04, 132.78]	76.97 [39.46, 160.40]	0.433	109.61 [76.04, 132.78]	105.87 [58.83, 160.75]	0.918	109.61 [76.04, 132.78]	101.63 [61.87, 160.26]	0.817
Sacin*VT, % of predicted normal value (median (IQR))	145.31 [138.98, 186.63]	125.81 [86.93, 171.41]	0.549	145.31 [138.98, 186.63]	116.32 [92.15, 169.75]	0.445	145.31 [138.98, 186.63]	113.92 [79.73, 172.62]	0.329
FEF50, % of predicted normal value (mean (SD))	79.88 (31.77)	72.29 (31.76)	0.515	79.88 (31.77)	66.34 (27.81)	0.190	79.88 (31.77)	62.89 (27.39)	0.097
Bronchodilator reversibility, % initial	8.98 (3.82)	11.46 (11.62)	0.550	8.98 (3.82)	13.41 (13.58)	0.361	8.98 (3.82)	14.46 (12.15)	0.208
Airway hyper-responsiveness, n(%)			0.780			0.576			0.917
Very mild	1 (14.3)	28 (29.5)		1 (14.3)	30 (36.1)		1 (14.3)	12 (15.6)	
Mild	2 (28.6)	30 (31.6)		2 (28.6)	21 (25.3)		2 (28.6)	21 (27.3)	
Moderate	3 (42.9)	28 (29.5)		3 (42.9)	19 (22.9)		3 (42.9)	25 (32.5)	
Severe	1 (14.3)	9 (9.5)		1 (14.3)	13 (15.7)		1 (14.3)	19 (24.7)	
	B	C	p	B	D	p	C	D	p
n	123	119		123	111		119	111	
Sex, male, n(%)	51 (41.5)	44 (37.0)	0.560	51 (41.5)	53 (47.7)	0.404	44 (37.0)	53 (47.7)	0.129
Age, years (mean (SD))	44.56 (12.83)	43.91 (13.06)	0.695	44.56 (12.83)	43.24 (13.55)	0.446	43.91 (13.06)	43.24 (13.55)	0.705
Smoking status, n(%)			0.333			0.570			0.060

Current smoker	5 (4.1)	9 (7.6)		5 (4.1)	2 (1.8)		9 (7.6)	2 (1.8)	
Ex smoker	28 (22.8)	32 (26.9)		28 (22.8)	24 (21.6)		32 (26.9)	24 (21.6)	
Non smoker	90 (73.2)	78 (65.5)		90 (73.2)	85 (76.6)		78 (65.5)	85 (76.6)	
BMI, kg/m2 (mean (SD))	26.54 (4.93)	26.97 (5.40)	0.518	26.54 (4.93)	27.60 (5.97)	0.139	26.97 (5.40)	27.60 (5.97)	0.404
Phadiatop test, Positive, n(%)	83 (76.1)	81 (87.1)	0.071	83 (76.1)	75 (83.3)	0.284	81 (87.1)	75 (83.3)	0.611
Duration of disease, years (median (IQR))	19.06 [7.59, 29.79]	20.11 [10.69, 30.38]	0.540	19.06 [7.59, 29.79]	21.67 [10.88, 32.48]	0.239	20.11 [10.69, 30.38]	21.67 [10.88, 32.48]	0.513
Age at first diagnosis, years (median (IQR))	23.00 [10.00, 38.47]	20.68 [6.79, 32.80]	0.229	23.00 [10.00, 38.47]	15.84 [5.59, 35.50]	0.039	20.68 [6.79, 32.80]	15.84 [5.59, 35.50]	0.331
FEV1, % of predicted normal value (mean (SD))	85.89 (16.84)	81.28 (17.27)	0.041	85.89 (16.84)	80.98 (17.53)	0.033	81.28 (17.27)	80.98 (17.53)	0.896
ACQ-6 total mean score (median (IQR))	0.67 [0.33, 1.17]	1.00 [0.33, 1.58]	0.078	0.67 [0.33, 1.17]	0.83 [0.33, 1.67]	0.082	1.00 [0.33, 1.58]	0.83 [0.33, 1.67]	0.979
Blood eosinophils, 10 ⁹ /L (median (IQR))	0.15 [0.10, 0.26]	0.20 [0.13, 0.30]	0.006	0.15 [0.10, 0.26]	0.29 [0.16, 0.40]	<0.001	0.20 [0.13, 0.30]	0.29 [0.16, 0.40]	0.002
Sputum lymphocytes, % of non-squamous cells (median (IQR))	0.80 [0.30, 1.65]	0.90 [0.45, 2.05]	0.137	0.80 [0.30, 1.65]	0.40 [0.30, 1.30]	0.344	0.90 [0.45, 2.05]	0.40 [0.30, 1.30]	0.010
Sputum eosinophils, % of non-squamous cells (median (IQR))	0.20 [0.00, 0.90]	0.50 [0.10, 2.95]	0.061	0.20 [0.00, 0.90]	1.20 [0.00, 6.50]	0.011	0.50 [0.10, 2.95]	1.20 [0.00, 6.50]	0.293
FeNO, ppb (median (IQR))	24.00 [15.00, 36.00]	21.00 [15.75, 30.75]	0.250	24.00 [15.00, 36.00]	29.00 [19.00, 51.00]	0.050	21.00 [15.75, 30.75]	29.00 [19.00, 51.00]	0.004
RV/TLC % of predicted normal value (mean (SD))	100.37 (24.12)	109.73 (28.25)	0.007	100.37 (24.12)	107.85 (27.46)	0.032	109.73 (28.25)	107.85 (27.46)	0.616
R5-20 % of predicted normal value (median (IQR))	200.21 [69.14, 669.36]	363.57 [135.94, 645.84]	0.108	200.21 [69.14, 669.36]	194.84 [73.21, 559.74]	0.744	363.57 [135.94, 645.84]	194.84 [73.21, 559.74]	0.042
Scnd*VT, % of predicted normal value (median (IQR))	76.97 [39.46, 160.40]	105.87 [58.83, 160.75]	0.116	76.97 [39.46, 160.40]	101.63 [61.87, 160.26]	0.181	105.87 [58.83, 160.75]	101.63 [61.87, 160.26]	0.754
Sacin*VT, % of predicted normal value (median (IQR))	125.81 [86.93, 171.41]	116.32 [92.15, 169.75]	0.824	125.81 [86.93, 171.41]	113.92 [79.73, 172.62]	0.388	116.32 [92.15, 169.75]	113.92 [79.73, 172.62]	0.515

FEF50, % of predicted normal value (mean (SD))	72.29 (31.76)	66.34 (27.81)	0.136	72.29 (31.76)	62.89 (27.39)	0.020	66.34 (27.81)	62.89 (27.39)	0.356
Bronchodilator reversibility, % initial	11.46 (11.62)	13.41 (13.58)	0.242	11.46 (11.62)	14.46 (12.15)	0.061	13.41 (13.58)	14.46 (12.15)	0.546
Airway hyper-responsiveness, n(%)			0.346			0.019			0.024
Very mild	28 (29.5)	30 (36.1)		28 (29.5)	12 (15.6)		30 (36.1)	12 (15.6)	
Mild	30 (31.6)	21 (25.3)		30 (31.6)	21 (27.3)		21 (25.3)	21 (27.3)	
Moderate	28 (29.5)	19 (22.9)		28 (29.5)	25 (32.5)		19 (22.9)	25 (32.5)	
Severe	9 (9.5)	13 (15.7)		9 (9.5)	19 (24.7)		13 (15.7)	19 (24.7)	

Supplementary Table 4. Pairwise comparison of asthma subgroups. **FEV₁**: Forced expiratory volume in 1 second, **FeNO**: fractional exhaled nitric oxide, **RV/TLC**: Residual volume/total lung capacity, **R5-20**: resistance at 5 Hz–resistance at 20 Hz, **Scond*VT**: ventilation heterogeneity in the conductive zone of the lungs, **Sacin*VT**: ventilation homogeneity of the acinar zone of the lungs, **FEF50**: Forced expiratory flow at 50% of predicted normal value. For normal data t-test was used, for non-normal data Mann-Whitney U test was performed.

Ontology	ID	Description	GeneRatio	BgRatio	pvalue	p.adjust	qvalue	Gene module
BP	GO:0030199	collagen fibril organization	5/17	64/17766	3.1e-09	1,00E-06	9,00E-07	1
BP	GO:0030198	extracellular matrix organization	7/17	319/17766	9.39e-09	1,00E-06	9,00E-07	1
BP	GO:0043062	extracellular structure organization	7/17	320/17766	9.6e-09	1,00E-06	9,00E-07	1
BP	GO:0045229	external encapsulating structure organization	7/17	321/17766	9.81e-09	1,00E-06	9,00E-07	1
BP	GO:0035987	endodermal cell differentiation	4/17	49/17766	1.18e-07	1,00E-05	9,00E-06	1
BP	GO:0007492	endoderm development	4/17	83/17766	1.01e-06	8,00E-05	5,00E-05	1
BP	GO:0032963	collagen metabolic process	4/17	102/17766	2.3e-06	2,00E-04	1,00E-04	1
BP	GO:0043200	response to amino acid	4/17	122/17766	4.7e-06	3,00E-04	2,00E-04	1
BP	GO:0016525	negative regulation of angiogenesis	4/17	126/17766	5.34e-06	3,00E-04	2,00E-04	1
BP	GO:2000181	negative regulation of blood vessel morphogenesis	4/17	128/17766	5.69e-06	3,00E-04	2,00E-04	1
BP	GO:1901343	negative regulation of vasculature development	4/17	129/17766	5.87e-06	3,00E-04	2,00E-04	1
BP	GO:0001101	response to acid chemical	4/17	136/17766	7.24e-06	3,00E-04	2,00E-04	1
BP	GO:0071604	transforming growth factor beta production	3/17	45/17766	1.01e-05	4,00E-04	2,00E-04	1
BP	GO:0071230	cellular response to amino acid stimulus	3/17	82/17766	6.15e-05	0.002	0.001	1
BP	GO:0071229	cellular response to acid chemical	3/17	90/17766	8.12e-05	0.003	0.002	1
BP	GO:0070206	protein trimerization	2/17	17/17766	0.000116	0.003	0.002	1
BP	GO:0006929	substrate-dependent cell migration	2/17	26/17766	0.000276	0.007	0.004	1
BP	GO:0018149	peptide cross-linking	2/17	28/17766	0.000321	0.008	0.005	1
BP	GO:0051897	positive regulation of phosphatidylinositol 3-kinase/protein kinase B signal transduction	3/17	165/17766	0.000486	0.01	0.007	1
BP	GO:0035909	aorta morphogenesis	2/17	36/17766	0.000533	0.01	0.007	1
BP	GO:0071634	regulation of transforming growth factor beta production	2/17	42/17766	0.000726	0.02	0.01	1

BP	GO:0030574	collagen catabolic process	2/17	44/17766	0.000796	0.02	0.01	1
BP	GO:0038084	vascular endothelial growth factor signaling pathway	2/17	50/17766	0.00103	0.02	0.01	1
BP	GO:0048008	platelet-derived growth factor receptor signaling pathway	2/17	57/17766	0.00133	0.02	0.01	1
BP	GO:0007566	embryo implantation	2/17	59/17766	0.00143	0.02	0.02	1
BP	GO:0008347	glial cell migration	2/17	60/17766	0.00148	0.02	0.02	1
BP	GO:0035904	aorta development	2/17	63/17766	0.00163	0.03	0.02	1
BP	GO:0014823	response to activity	2/17	69/17766	0.00195	0.03	0.02	1
BP	GO:0035924	cellular response to vascular endothelial growth factor stimulus	2/17	71/17766	0.00206	0.03	0.02	1
BP	GO:0071560	cellular response to transforming growth factor beta stimulus	3/17	275/17766	0.00212	0.03	0.02	1
BP	GO:0071559	response to transforming growth factor beta	3/17	281/17766	0.00226	0.03	0.02	1
BP	GO:0061448	connective tissue development	3/17	282/17766	0.00228	0.03	0.02	1
BP	GO:0043588	skin development	3/17	313/17766	0.00307	0.04	0.03	1
BP	GO:0048661	positive regulation of smooth muscle cell proliferation	2/17	90/17766	0.00328	0.04	0.03	1
CC	GO:0062023	collagen-containing extracellular matrix	12/18	422/18712	2.44e-16	1,00E-14	6,00E-15	1
CC	GO:0005581	collagen trimer	6/18	86/18712	1.4e-10	2,00E-09	1,00E-09	1
CC	GO:0005788	endoplasmic reticulum lumen	6/18	310/18712	3.09e-07	2,00E-06	1,00E-06	1
CC	GO:0031091	platelet alpha granule	3/18	91/18712	8.61e-05	4,00E-04	2,00E-04	1
CC	GO:0016324	apical plasma membrane	3/18	402/18712	0.00632	0.02	0.01	1
CC	GO:0045177	apical part of cell	3/18	468/18712	0.00959	0.03	0.02	1
CC	GO:0031256	leading edge membrane	2/18	179/18712	0.0126	0.04	0.02	1
CC	GO:0030673	axolemma	1/18	15/18712	0.0143	0.04	0.02	1
CC	GO:0097449	astrocyte projection	1/18	19/18712	0.0181	0.05	0.03	1
MF	GO:0005201	extracellular matrix structural constituent	10/18	165/17746	1.5e-16	9,00E-15	5,00E-15	1

MF	GO:0048407	platelet-derived growth factor binding	4/18	11/17746	2.43e-10	8,00E-09	4,00E-09	1
MF	GO:0005518	collagen binding	3/18	68/17746	4.21e-05	5,00E-04	3,00E-04	1
MF	GO:0005539	glycosaminoglycan binding	4/18	233/17746	7.67e-05	8,00E-04	4,00E-04	1
MF	GO:0002020	protease binding	3/18	138/17746	0.000345	0.003	0.001	1
MF	GO:0005178	integrin binding	3/18	154/17746	0.000475	0.003	0.002	1
MF	GO:0043394	proteoglycan binding	2/18	36/17746	6,00E-04	0.004	0.002	1
MF	GO:0050840	extracellular matrix binding	2/18	54/17746	0.00135	0.007	0.003	1
MF	GO:1901681	sulfur compound binding	3/18	269/17746	0.00237	0.01	0.006	1
MF	GO:0046332	SMAD binding	2/18	79/17746	0.00286	0.01	0.006	1
MF	GO:0004222	metalloendopeptidase activity	2/18	107/17746	0.00517	0.02	0.01	1
MF	GO:0000099	sulfur amino acid transmembrane transporter activity	1/18	11/17746	0.0111	0.04	0.02	1
MF	GO:0004859	phospholipase inhibitor activity	1/18	12/17746	0.0121	0.04	0.02	1
MF	GO:0004252	serine-type endopeptidase activity	2/18	168/17746	0.0123	0.04	0.02	1
MF	GO:0005161	platelet-derived growth factor receptor binding	1/18	14/17746	0.0141	0.04	0.02	1
MF	GO:0005313	L-glutamate transmembrane transporter activity	1/18	14/17746	0.0141	0.04	0.02	1
MF	GO:0008236	serine-type peptidase activity	2/18	186/17746	0.015	0.04	0.02	1
MF	GO:0055102	lipase inhibitor activity	1/18	15/17746	0.0151	0.04	0.02	1
MF	GO:0017171	serine hydrolase activity	2/18	190/17746	0.0156	0.04	0.02	1
MF	GO:0015172	acidic amino acid transmembrane transporter activity	1/18	18/17746	0.0181	0.04	0.02	1
MF	GO:0061134	peptidase regulator activity	2/18	221/17746	0.0207	0.05	0.02	1
BP	GO:0008544	epidermis development	4/7	380/17766	6.85e-06	0.001	5,00E-04	2
BP	GO:0043588	skin development	3/7	313/17766	0.00018	0.01	0.006	2
BP	GO:0045109	intermediate filament organization	2/7	73/17766	0.000345	0.02	0.008	2

BP	GO:0045103	intermediate filament-based process	2/7	95/17766	0.000584	0.02	0.008	2
BP	GO:0042157	lipoprotein metabolic process	2/7	133/17766	0.00114	0.03	0.01	2
BP	GO:0030216	keratinocyte differentiation	2/7	173/17766	0.00192	0.04	0.02	2
CC	GO:0001533	cornified envelope	3/7	60/18712	1.09e-06	4,00E-05	2,00E-05	2
CC	GO:0045095	keratin filament	2/7	77/18712	0.000346	0.006	0.003	2
CC	GO:0031527	filopodium membrane	1/7	18/18712	0.00672	0.04	0.02	2
CC	GO:0031258	lamellipodium membrane	1/7	22/18712	0.0082	0.05	0.02	2
CC	GO:0030055	cell-substrate junction	2/7	430/18712	0.0102	0.05	0.02	2
CC	GO:0008305	integrin complex	1/7	31/18712	0.0115	0.05	0.02	2
CC	GO:0031528	microvillus membrane	1/7	33/18712	0.0123	0.05	0.02	2
MF	GO:0031994	insulin-like growth factor I binding	2/7	13/17746	1.04e-05	5,00E-04	3,00E-04	2
MF	GO:0001968	fibronectin binding	2/7	29/17746	5.39e-05	9,00E-04	5,00E-04	2
MF	GO:0019215	intermediate filament binding	1/7	11/17746	0.00433	0.04	0.03	2
BP	GO:0002181	cytoplasmic translation	9/33	154/17766	7.05e-12	4,00E-09	3,00E-09	3
BP	GO:1903311	regulation of mRNA metabolic process	6/33	324/17766	2.57e-05	0.005	0.004	3
BP	GO:1903312	negative regulation of mRNA metabolic process	4/33	94/17766	2.67e-05	0.005	0.004	3
BP	GO:0070934	CRD-mediated mRNA stabilization	2/33	11/17766	0.000182	0.01	0.01	3
BP	GO:0008380	RNA splicing	6/33	464/17766	0.000187	0.01	0.01	3
BP	GO:1900152	negative regulation of nuclear-transcribed mRNA catabolic process, deadenylation-dependent decay	2/33	12/17766	0.000218	0.01	0.01	3
BP	GO:0022613	ribonucleoprotein complex biogenesis	6/33	490/17766	0.000251	0.01	0.01	3
BP	GO:0000288	nuclear-transcribed mRNA catabolic process, deadenylation-dependent decay	3/33	69/17766	0.000281	0.01	0.01	3

BP	GO:0042254	ribosome biogenesis	5/33	319/17766	0.000284	0.01	0.01	3
BP	GO:2000767	positive regulation of cytoplasmic translation	2/33	14/17766	3,00E-04	0.01	0.01	3
BP	GO:0000281	mitotic cytokinesis	3/33	87/17766	0.000556	0.02	0.02	3
BP	GO:0090398	cellular senescence	3/33	102/17766	0.000885	0.02	0.02	3
BP	GO:0031111	negative regulation of microtubule polymerization or depolymerization	2/33	38/17766	0.00226	0.04	0.03	3
BP	GO:0007019	microtubule depolymerization	2/33	45/17766	0.00315	0.04	0.03	3
BP	GO:0034250	positive regulation of amide metabolic process	3/33	163/17766	0.00338	0.05	0.03	3
BP	GO:2000772	regulation of cellular senescence	2/33	47/17766	0.00343	0.05	0.03	3
CC	GO:0022625	cytosolic large ribosomal subunit	8/34	58/18712	8.79e-14	1,00E-11	8,00E-12	3
CC	GO:0005925	focal adhesion	6/34	420/18712	9.74e-05	0.001	9,00E-04	3
CC	GO:0005844	polysome	3/34	67/18712	0.000243	0.003	0.002	3
CC	GO:0005681	spliceosomal complex	4/34	197/18712	0.000432	0.004	0.003	3
CC	GO:0005874	microtubule	5/34	464/18712	0.00141	0.01	0.008	3
CC	GO:0002102	podosome	2/34	31/18712	0.00144	0.01	0.008	3
CC	GO:0000922	spindle pole	3/34	175/18712	0.00389	0.03	0.02	3
CC	GO:0030027	lamellipodium	3/34	201/18712	0.00572	0.04	0.03	3
MF	GO:0003735	structural constituent of ribosome	7/31	164/17746	1.1e-08	1,00E-06	7,00E-07	3
MF	GO:0090079	translation regulator activity, nucleic acid binding	3/31	103/17746	0.000758	0.04	0.02	3
MF	GO:0051082	unfolded protein binding	3/31	115/17746	0.00104	0.04	0.02	3
MF	GO:0140693	molecular condensate scaffold activity	2/31	32/17746	0.00142	0.04	0.02	3
MF	GO:0140662	ATP-dependent protein folding chaperone	2/31	35/17746	0.0017	0.04	0.02	3
MF	GO:0045296	cadherin binding	4/31	333/17746	0.00257	0.04	0.03	3
MF	GO:0004869	cysteine-type endopeptidase inhibitor activity	2/14	34/17746	0.00032	0.03	0.02	4

Ontology	ID	Description	GeneRatio	BgRatio	pvalue	p.adjust	qvalue	
BP	GO:0051412	response to corticosterone	2/7	14/17766	1.21e-05	0.002	0.001	6
BP	GO:0071248	cellular response to metal ion	3/7	193/17766	4.28e-05	0.003	0.001	6
BP	GO:0071241	cellular response to inorganic substance	3/7	220/17766	6.32e-05	0.003	0.001	6
BP	GO:0071276	cellular response to cadmium ion	2/7	33/17766	6.99e-05	0.003	0.001	6
BP	GO:0051385	response to mineralocorticoid	2/7	37/17766	8.8e-05	0.003	0.001	6
BP	GO:0032570	response to progesterone	2/7	39/17766	9.79e-05	0.003	0.001	6
BP	GO:0140467	integrated stress response signaling	2/7	42/17766	0.000114	0.003	0.001	6
BP	GO:1902895	positive regulation of miRNA transcription	2/7	51/17766	0.000168	0.004	0.002	6
BP	GO:0046686	response to cadmium ion	2/7	54/17766	0.000189	0.004	0.002	6
BP	GO:2000630	positive regulation of miRNA metabolic process	2/7	61/17766	0.000241	0.004	0.002	6
BP	GO:0061614	miRNA transcription	2/7	70/17766	0.000317	0.005	0.002	6
BP	GO:0060395	SMAD protein signal transduction	2/7	79/17766	0.000404	0.005	0.002	6
BP	GO:0051591	response to cAMP	2/7	84/17766	0.000457	0.005	0.002	6
BP	GO:0140747	regulation of ncRNA transcription	2/7	91/17766	0.000536	0.006	0.002	6
BP	GO:0010586	miRNA metabolic process	2/7	102/17766	0.000673	0.007	0.003	6
BP	GO:0046683	response to organophosphorus	2/7	126/17766	0.00102	0.009	0.004	6
BP	GO:0034614	cellular response to reactive oxygen species	2/7	145/17766	0.00135	0.01	0.004	6
BP	GO:0007565	female pregnancy	2/7	184/17766	0.00217	0.02	0.006	6
BP	GO:0000302	response to reactive oxygen species	2/7	196/17766	0.00245	0.02	0.007	6
BP	GO:1902075	cellular response to salt	2/7	196/17766	0.00245	0.02	0.007	6
BP	GO:1901654	response to ketone	2/7	205/17766	0.00268	0.02	0.007	6
BP	GO:0009612	response to mechanical stimulus	2/7	212/17766	0.00286	0.02	0.007	6
BP	GO:0044706	multi-multicellular organism process	2/7	212/17766	0.00286	0.02	0.007	6

BP	GO:0007179	transforming growth factor beta receptor signaling pathway	2/7	219/17766	0.00305	0.02	0.007	6
BP	GO:0033002	muscle cell proliferation	2/7	225/17766	0.00322	0.02	0.007	6
BP	GO:0015820	leucine transport	1/7	10/17766	0.00393	0.02	0.008	6
BP	GO:0071374	cellular response to parathyroid hormone stimulus	1/7	10/17766	0.00393	0.02	0.008	6
BP	GO:0000101	sulfur amino acid transport	1/7	12/17766	0.00472	0.02	0.009	6
BP	GO:0071559	response to transforming growth factor beta	2/7	281/17766	0.00497	0.02	0.01	6
BP	GO:0071107	response to parathyroid hormone	1/7	13/17766	0.00511	0.02	0.01	6
BP	GO:0043922	negative regulation by host of viral transcription	1/7	14/17766	0.0055	0.02	0.01	6
BP	GO:0010958	regulation of amino acid import across plasma membrane	1/7	16/17766	0.00629	0.02	0.01	6
BP	GO:0051956	negative regulation of amino acid transport	1/7	16/17766	0.00629	0.02	0.01	6
BP	GO:1903789	regulation of amino acid transmembrane transport	1/7	16/17766	0.00629	0.02	0.01	6
BP	GO:0039535	regulation of RIG-I signaling pathway	1/7	20/17766	0.00785	0.03	0.01	6
BP	GO:1902074	response to salt	2/7	360/17766	0.00804	0.03	0.01	6
BP	GO:0035902	response to immobilization stress	1/7	23/17766	0.00903	0.03	0.01	6
BP	GO:0006979	response to oxidative stress	2/7	390/17766	0.00938	0.03	0.01	6
BP	GO:0014856	skeletal muscle cell proliferation	1/7	24/17766	0.00942	0.03	0.01	6
BP	GO:0045672	positive regulation of osteoclast differentiation	1/7	27/17766	0.0106	0.03	0.01	6
BP	GO:0009410	response to xenobiotic stimulus	2/7	425/17766	0.0111	0.03	0.01	6
BP	GO:0015697	quaternary ammonium group transport	1/7	29/17766	0.0114	0.03	0.01	6
BP	GO:0051450	myoblast proliferation	1/7	29/17766	0.0114	0.03	0.01	6
BP	GO:0043392	negative regulation of DNA binding	1/7	35/17766	0.0137	0.04	0.02	6

BP	GO:0039532	negative regulation of viral-induced cytoplasmic pattern recognition receptor signaling pathway	1/7	36/17766	0.0141	0.04	0.02	6
BP	GO:1904707	positive regulation of vascular associated smooth muscle cell proliferation	1/7	44/17766	0.0172	0.05	0.02	6
BP	GO:0006305	DNA alkylation	1/7	46/17766	0.018	0.05	0.02	6
BP	GO:0006306	DNA methylation	1/7	46/17766	0.018	0.05	0.02	6
BP	GO:0019083	viral transcription	1/7	51/17766	0.0199	0.05	0.02	6
CC	GO:0090575	RNA polymerase II transcription regulator complex	2/7	244/18712	0.00341	0.02	0.01	6
CC	GO:0005667	transcription regulator complex	2/7	499/18712	0.0136	0.05	0.03	6
MF	GO:0070412	R-SMAD binding	2/7	25/17746	3.98e-05	0.002	6,00E-04	6
MF	GO:0001228	DNA-binding transcription activator activity, RNA polymerase II-specific	3/7	461/17746	0.000564	0.006	0.002	6
MF	GO:0001216	DNA-binding transcription activator activity	3/7	465/17746	0.000578	0.006	0.002	6
MF	GO:0015173	aromatic amino acid transmembrane transporter activity	1/7	10/17746	0.00394	0.02	0.009	6
MF	GO:0015658	branched-chain amino acid transmembrane transporter activity	1/7	10/17746	0.00394	0.02	0.009	6
MF	GO:0000099	sulfur amino acid transmembrane transporter activity	1/7	11/17746	0.00433	0.02	0.009	6
MF	GO:0035497	cAMP response element binding	1/7	12/17746	0.00472	0.02	0.009	6
MF	GO:0015926	glucosidase activity	1/7	14/17746	0.00551	0.03	0.01	6
MF	GO:0005402	carbohydrate:monoatomic cation symporter activity	1/7	16/17746	0.0063	0.03	0.01	6
MF	GO:0061629	RNA polymerase II-specific DNA-binding transcription factor binding	2/7	343/17746	0.00734	0.03	0.01	6
MF	GO:0000979	RNA polymerase II core promoter sequence-specific DNA binding	1/7	19/17746	0.00747	0.03	0.01	6
MF	GO:0140311	protein sequestering activity	1/7	23/17746	0.00904	0.03	0.01	6
MF	GO:0015295	solute:proton symporter activity	1/7	24/17746	0.00943	0.03	0.01	6

MF	GO:0015144	carbohydrate transmembrane transporter activity	1/7	42/17746	0.0165	0.04	0.01	6
MF	GO:0140296	general transcription initiation factor binding	1/7	54/17746	0.0211	0.04	0.02	6
MF	GO:1901682	sulfur compound transmembrane transporter activity	1/7	58/17746	0.0227	0.05	0.02	6

Supplementary Table 5. Gene ontology (GO) overrepresentation analysis for 6 gene modules. Analysis was performed using enrichGO function from the clusterProfiler package. Simplify function was used to remove redundant GO terms from the enrichment result.

Gene	Gene symbol	logFC	PValue	FDR
ENSG00000117983	MUC5B	-1.2	0.00014	0.0023
ENSG00000145113	MUC4	0.26	0.0036	0.031
ENSG00000185499	MUC1	0.27	0.018	0.091
ENSG00000261272	MUC22	0.73	0.025	0.091
ENSG00000181143	MUC16	0.18	0.027	0.091
ENSG00000204544	MUC21	0.3	0.05	0.14
ENSG00000169550	MUC15	-0.21	0.1	0.25
ENSG00000242086	MUC20-OT1	0.14	0.17	0.35
ENSG00000173702	MUC13	0.12	0.53	0.82
ENSG00000176945	MUC20	0.042	0.58	0.82
ENSG00000184956	MUC6	-0.2	0.5	0.82
ENSG00000172551	MUCL1	-0.11	0.46	0.82
ENSG00000215182	MUC5AC	0.051	0.68	0.83
ENSG00000205277	MUC12	0.073	0.73	0.83
ENSG00000227053	MUC12-AS1	-0.094	0.69	0.83
ENSG00000198788	MUC2	-0.045	0.86	0.91
ENSG00000224769	MUC20P1	-0.014	0.94	0.94

Supplementary Table 6. Mucins differential gene expression between asthma subgroup C and healthy samples.